inClass-L37

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inClass-L37

CV2090
Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

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L-37 (2/4; risk-n-rel)
Oct 26, 2026 (2 PM)

The R-S Reliability Problem

Finding p_f

Limit State Function

R : Capacity

S : Demand

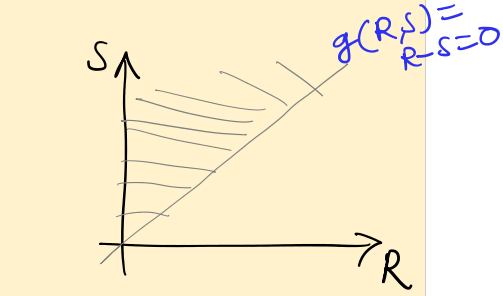
Limit State function

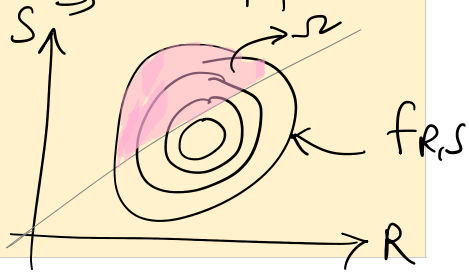
$g(R,S) = R - S$

$R - S > 0 \Rightarrow R > S \Rightarrow$ Safe

$R - S < 0 \Rightarrow$ Unsafe / failure

$$p_f = \int_{\Omega} f_{R,S}(r,s)\,dr\,ds$$





An Example: Structural Column

Demand RVs: D, L, W, E

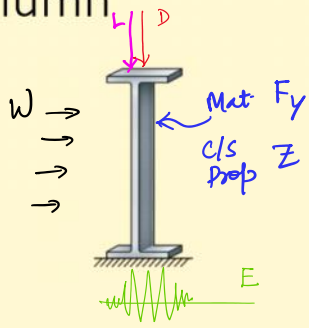
Capacity RVs: F_y, Z

Limit state

$g(\ ) = F_y Z - D - L - W - E$

Objective:

$Pr[g(\ ) < 0]$



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The R-S reliability problem

$p_f = Pr[R - S < 0]$

Let's assume that capacity is discrete,

$R = r_i$

$Pr[\text{Failure} \cap R = r_i] \leftarrow p_{f_i}$

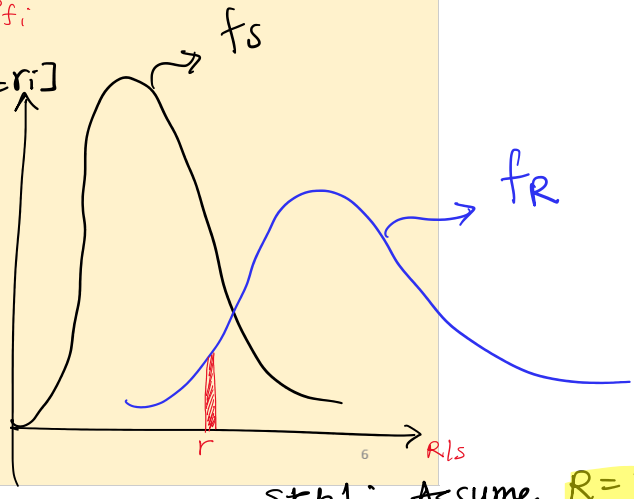
$= Pr[R = r_i] \cdot Pr[\text{Failure} \mid R = r_i]$

$p_f = \sum_i p_{f_i}$

$= \sum Pr[R = r_i] \cdot Pr[S > R \mid R = r_i]$

↳ Generalize p_f

When R & S are both Continuous RV



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step 1: Assume $R = r$

step 2: Conditional prob of failure $\int_r^\infty f_S\, ds$

step 3: Total prob. Rule $p_f = \int_0^\infty f_R(r) \left[\int_r^\infty f_S\, ds \right] \cdot dr$

The R-S reliability problem

$p_f = Pr(R < S)$

$=$

Conditioned over R

$$p_f = \int_0^\infty f_R(r) \cdot [1 - F_S(r)]\, dr$$

Conditioned over S

Conditioned over S

$$p_f = \int_0^\infty f_S(s)\, F_R(s)\, ds$$

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The R-S Problem: Safety Margin

$g(R,S) = R - S$

Capacity, Demand

Safety Factor $g(R,S) = \frac{R}{S} - 1$

The R-S reliability problem

- Formulation in terms of safety margin

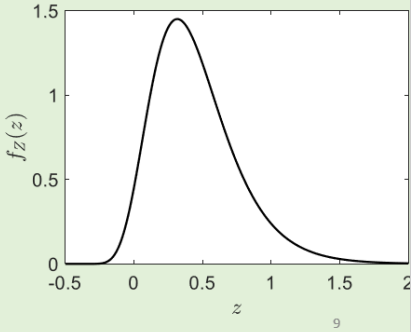
A new RV $Z = R - S$

$p_f = Pr(Z < 0)$

If R & S are jointly Normal

$\mu_Z = \mu_R - \mu_S$

$\sigma_Z^2 = \sigma_R^2 + \sigma_S^2 - 2\rho_{RS}\sigma_R\sigma_S$



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The R-S reliability problem

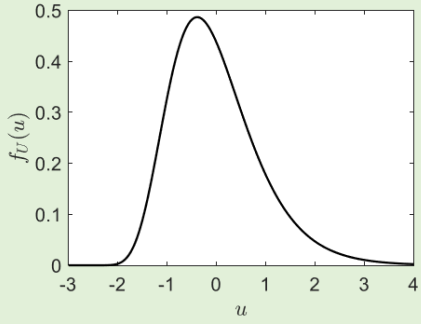
- Formulation in terms of safety margin

$Z = \quad R - S$

$p_f = \quad \Pr(Z \leq 0) = F_Z(0)$

$U =$

$p_f =$



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The R-S reliability problem

- Special case: R and S are jointly normal (ok if small c.o.v.)

$Z = \mathrm{N}(\mu_Z, \sigma_Z), \qquad U = \mathrm{N}(0,1)$

$p_f = \Pr(Z \leq 0) = F_U(-\beta) = \Phi\,(-\beta)$

$\beta =$

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The R-S Problem: Safety Factor

$$g(R,S) = \frac{R}{S} - 1$$

The R-S reliability problem

- Formulation in terms of (logarithmic) safety factor

$$Z = \ln R - \ln S$$

$$\mu_Z \cong \ln \mu_R - \ln \mu_S - \frac{1}{2}(\delta_R^2 - \delta_S^2)$$

$$\sigma_Z^2 \cong \delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2$$

$$\beta \cong \frac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2}}$$

In terms of: $r, \delta_R, \delta_S, \rho_{RS}$

An approximation is to assume $U = \mathrm{N}(0,1)$. Then $p_f \cong \Phi(-\beta)$

The R-S reliability problem

- Special case: R and S are jointly lognormal

$$\mu_Z = \ln\left(r\sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}}\right),$$

$$\sigma_Z^2 = \ln\left[\frac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}\right]$$

$$\beta = \frac{\mu_Z}{\sigma_Z} = \frac{\ln\left(r\sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}}\right)}{\sqrt{\ln\left[\frac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}\right]}}, \qquad p_f = \Phi(-\beta)$$

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The R-S reliability problem

Formulation	Reliability Index β	Remark
Safety margin $Z = R - S$	$\beta = \mu_z/\sigma_z$	General definition
R and S are jointly normal	$\beta = \dfrac{r-1}{\sqrt{r^2\delta_R^2 - 2r\rho_{RS}\delta_R\delta_S + \delta_S^2}}$	Exact
Safety factor $Z = \ln R - \ln S$	SO approx. $\beta \cong \dfrac{\ln r - \frac{1}{2}(\delta_R^2 - \delta_S^2)}{\sqrt{\delta_R^2 - 2\rho_{RS}\delta_R\delta_S + \delta_S^2}}$	Approximation but general
R and S are jointly lognormal	$\beta = \dfrac{\ln\left(r\sqrt{\frac{1+\delta_S^2}{1+\delta_R^2}}\right)}{\sqrt{\ln\left[\frac{(1+\delta_R^2)(1+\delta_S^2)}{(1+\rho_{RS}\delta_R\delta_S)^2}\right]}}$	Exact but special

$p_f \cong \Phi(-\beta)$. Every row requires only four number $r, \delta_R, \delta_S, \rho_{RS}$.
Every row assumes the shape of the distribution.

Monte Carlo needs neither.

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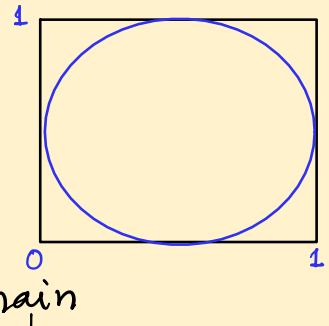
Monte Carlo Simulations

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Monte Carlo simulation

Integration using sampling

Step 1:
Throw N points randomly in a unit square.



main

- Throw points in domain
  - Generate N points between (0, 1)
  - Find the fraction of points within the circle. (0, 1)
- Estimate the running mean

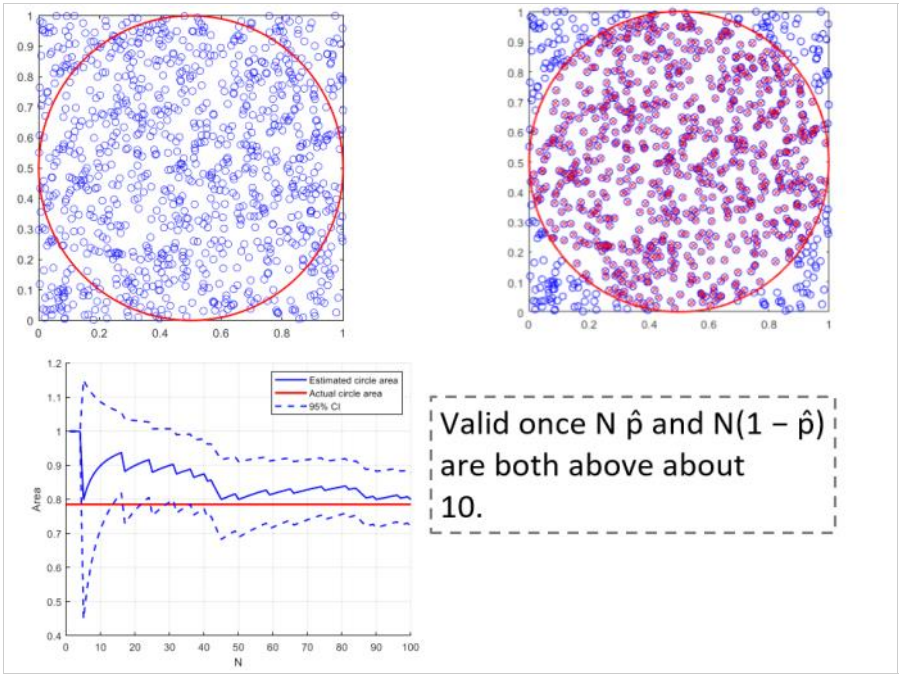
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Monte Carlo Simulations
Circle Area

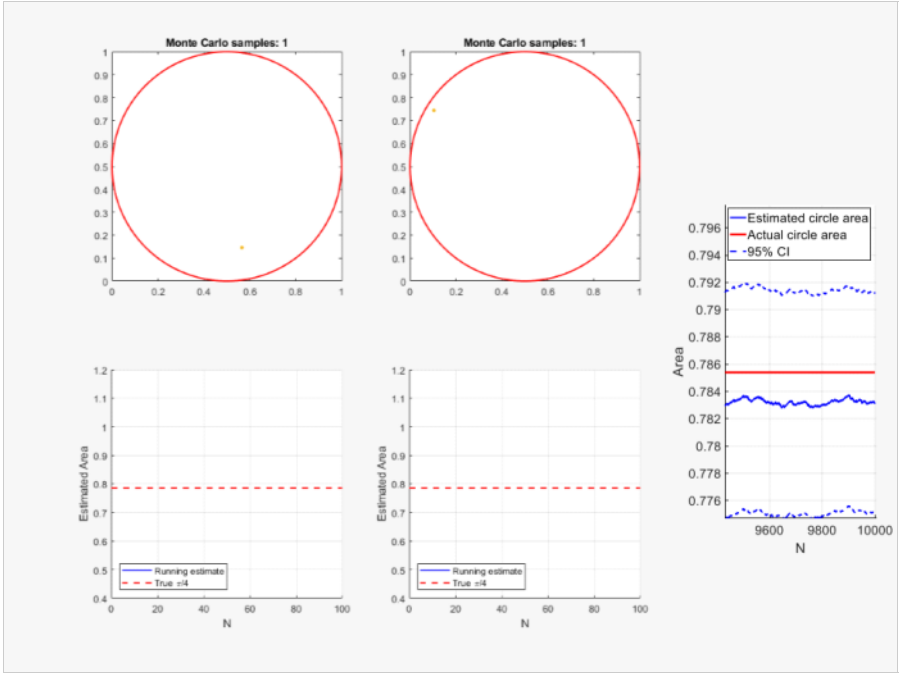
MCS Algorithm

Things you will need:

```
import numpy as np
rng = np.random.default_rng(2026)    # a seeded random number generator
rng.random(N)                        # N numbers, uniform on [0, 1]
rng.normal(mu, sigma, N)             # N numbers, normal
np.cumsum(q)           # gives the running total. Avoids loop
```



Valid once $N\hat{p}$ and $N(1-\hat{p})$ are both above about 10.



MC estimator properties

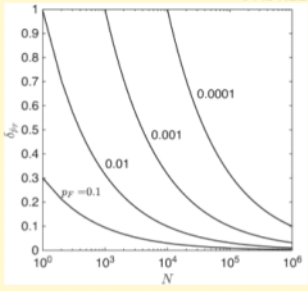
MC estimator

- Expected values, $\mathrm{E}[\hat{p}_F]$
- Variance, $\mathrm{Var}[\hat{p}_F]$
- c.o.v.

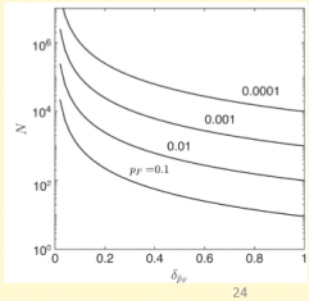
MC estimator properties

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- c.o.v.



- Sample required



MCS computer simulation

Computer implementation

$\delta_{\hat{p}_F} = \sqrt{\frac{1-p_F}{Np_F}}$ does not work, since p_F is unknown

- Keep two running sums $\sum_{i=1}^{N} q_i$ and $\sum_{i=1}^{N} q_i^2$.
- At any N, compute

$\bar{q}(N) = \frac{1}{N}\sum_{i=1}^{N} q_i$ $\qquad$ $\hat{p}_F(N) \cong \bar{q}(N)$

$se_q^2(N) = \frac{1}{N}\sum_{i=1}^{N} q_i^2 - \bar{q}^2$

$\hat{\delta}(N) \cong \frac{se_q(N)}{\sqrt{N}\bar{q}(N)}$

- Stop when $\hat{\delta}(N) \leq \delta_0$ or $N_0 \leq N$ for specified thresholds δ_0 and N_0.
- Always report $\hat{p}_F(N)$, $\hat{\delta}_{\hat{p}_F}(N)$, and N.

Monte Carlo Simulations

Finding p_f

Finding p_f by simulation: two problems

- **Case A**

$$g = R - S$$

$$R \sim \mathcal{N}(200, 20)\ \text{kNm}; \qquad S \sim \mathcal{N}(130, 25)\ \text{kNm}$$

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Finding p_f by simulation: two problems

- **Case B**

$$g = f_y Z - (D + L + W)$$

Variable	Distribution	Mean	s.d.	Units
F_y	Normal	250	25	MPa
Z	Normal	1.2	0.06	10^6 mm^3
D	Normal	100	10	kN-m
L	Normal	60	18	kN-m
W	Normal	40	16	kN-m

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MCS for p_f

- **Case B**

$g = f_y Z - (D + L + W)$

Variable	Distribution	Mean	s.d.	Units
F_y	Normal	250	25	MPa
Z	Normal	1.2	0.06	10^6 mm^3
D	Normal	100	10	kN-m
L	Normal	60	18	kN-m
W	Normal	40	16	kN-m

Step 1. define $g(X)$, and the distribution of every variable in X

Step 2. draw N samples of X rng.normal(mu, sigma, N)

Step 3. $q_i = 1$ if $g(x_i)$ <= 0, else 0

Step 4. $\hat{p}_f$ = mean(q); δ = running c. o. v. np.cumsum

Step 5. plot $\hat{p}_f$ and δ against N; report $\hat{p}_F(N)$, $\hat{\delta}_{\hat{p}_F}(N)$,and N

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MCS for p_f

- **Case B**

$g = f_y Z - (D + L + W)$

Variable	Distribution	Mean	s.d.	Units
F_y	Normal	250	25	MPa
Z	Normal	1.2	0.06	10^6 mm^3
D	Normal	100	10	kN-m
L	Normal	60	18	kN-m
W	Normal	40	16	kN-m

Result, N = 100,000, seed 2026:

	$\hat{p}_f$	$\hat{\delta}$
Case A (exact 0.01439)	0.01448	0.026
Case B (no exact value)	0.00803	0.035

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Monte Carlo Simulations
General integration

MCS for Integration

$$I = \int_{a_1}^{b_1}\int_{a_2}^{b_2}\cdots\int_{a_5}^{b_5} h(x)\, dx_1\, dx_2\, dx_3\, dx_4\, dx_5$$

Four dimensions, same 3 steps

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Thank you!

Hazard Functions (brief)

Hazard function

- In classical reliability theory, Hazard function is defined as:

Probability of failure in the next unit of time:

$$h(t)\mathrm{d}t = \Pr(t < T \leq t + \mathrm{d}t \mid t < T)$$

$$= \frac{\Pr(t < T \leq t + \mathrm{d}t)}{\Pr(t < T)}$$

$$= \frac{f_T(t)\mathrm{d}t}{1 - F_T(t)}$$

Hazard function

$$\Rightarrow h(t) = \frac{f_T(t)}{1 - F_T(t)}$$

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